31_Data_Visualization

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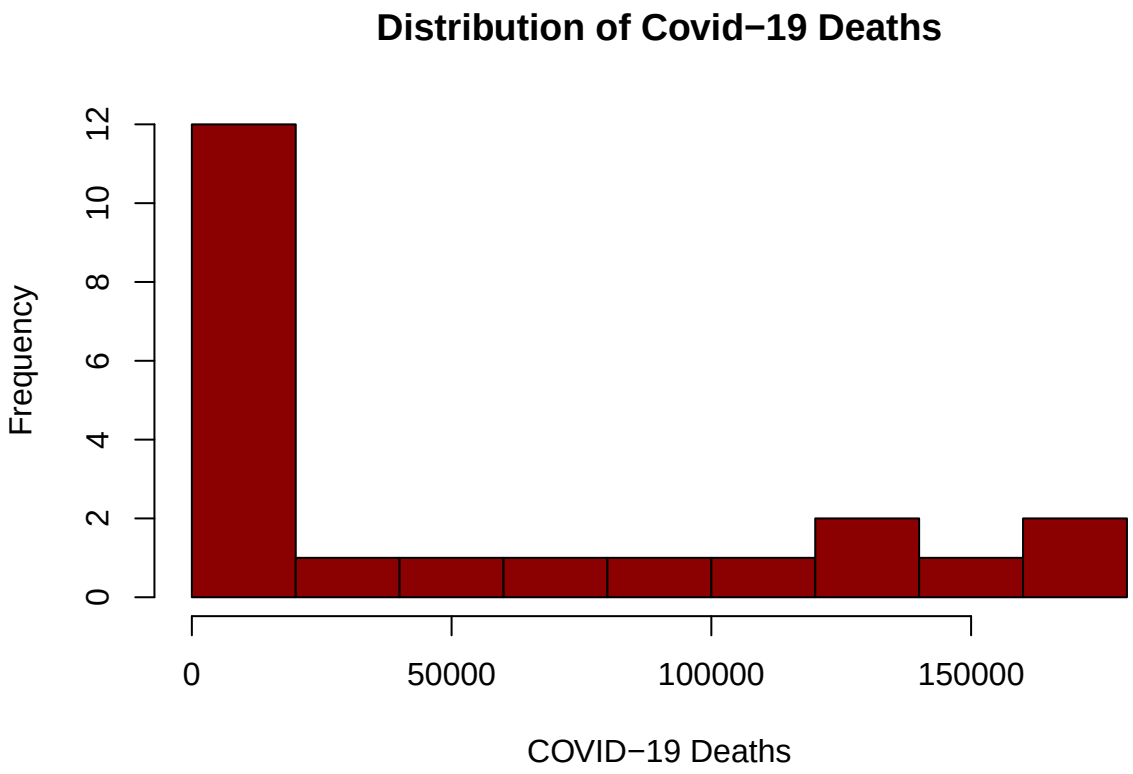
Loading the Clean Dataset

```
load("22_DataSaving.RData")
head(covid_cleaned)
```

```
##     Sex    Age.Group COVID.19.Deaths Total.Deaths Pneumonia.Deaths
## 19 Male Under 1 year             284        40371              586
## 21 Male    1-4 years             151         8102              341
## 22 Male   5-14 years             261        12850              449
## 23 Male  15-24 years            1797        97748             1917
## 25 Male  25-34 years            7607       196824             7129
## 27 Male  35-44 years           18602       273868            16415
##    Pneumonia.and.COVID.19.Deaths covid_pct_of_all pneu_pct_of_all
## 19                            54        0.7034753        1.451537
## 21                            35        1.8637373        4.208837
## 22                            83        2.0311284        3.494163
## 23                           730        1.8384008        1.961165
## 25                          3594        3.8648742        3.622018
## 27                          9399        6.7923233        5.993763
##    pneu_pct_of_covid age_mid
## 19          19.01408     0.5
## 21          23.17881     2.5
## 22          31.80077    10.0
## 23          40.62326    20.0
## 25          47.24596    30.0
## 27          50.52683    40.0
```

Plot 1: Histogram of COVID-19 Death Distribution

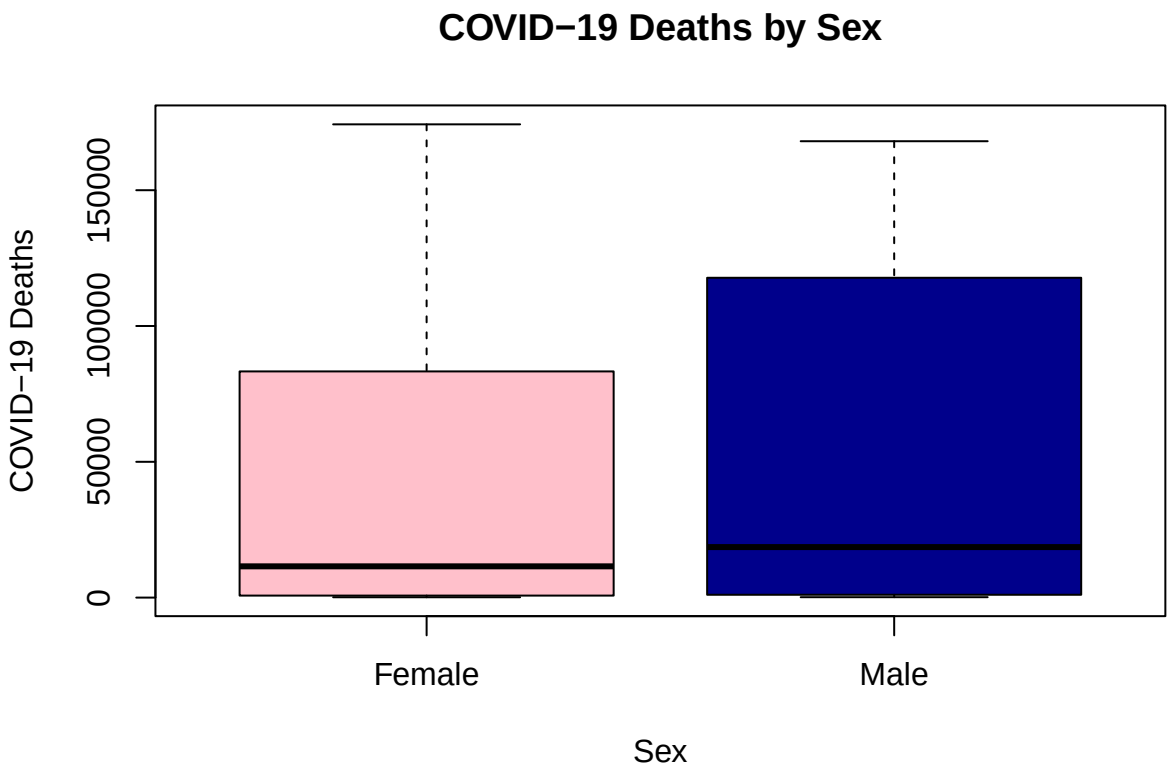
```
hist(covid_cleaned$COVID.19.Deaths,
    main = "Distribution of Covid-19 Deaths",
    xlab = "COVID-19 Deaths",
    col = 'darkred',
    breaks = 10)
```



This histogram displays the overall distribution and frequency of COVID-19 Deaths counts across all filters. It highlights that the lower death counts show up as small parts whereas the high deaths show up as a outlier.

Plot 2: Boxplot of COVID-19 Deaths by Sex

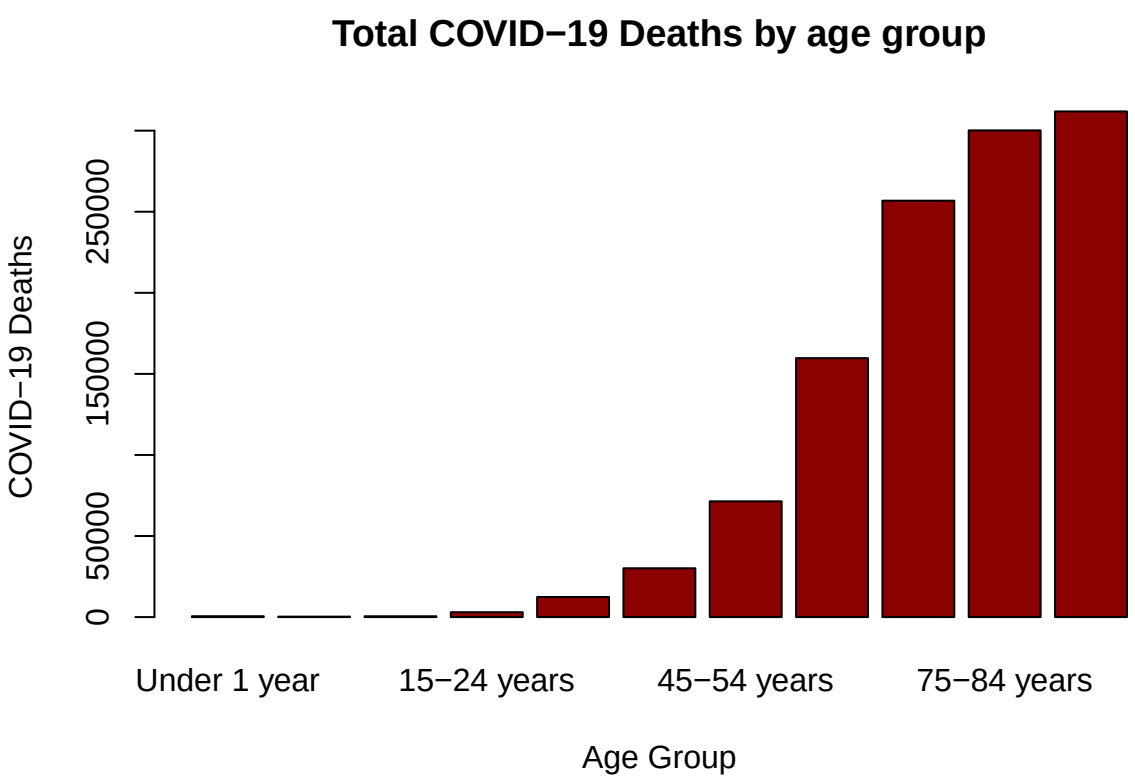
```
boxplot(COVID.19.Deaths ~Sex,
        data = covid_cleaned,
        main = "COVID-19 Deaths by Sex",
        xlab = "Sex",
        ylab = 'COVID-19 Deaths',
        col = c('pink','darkblue'))
```



This box plot compares the median, spread and range of COVID-19 deaths between males and females. It shows variability and highlights whether one group had a higher mortality rate.

Plot 3: Bar Plot of COVID-19 deaths by age

```
age_totals <- tapply(covid_cleaned$COVID.19.Deaths, covid_cleaned$Age.Group, sum)
barplot(age_totals,
        main = 'Total COVID-19 Deaths by age group',
        xlab = 'Age Group',
        ylab = 'COVID-19 Deaths',
        col = 'darkred')
```



This bar plot shows the total COVID-19 Deaths across age groups only. We are clearly able to intrepret that as someone gets older, they have a higher mortality rate and are more prone to be affected by COVID-19

Plot 4: Line plot for percentage of all deaths

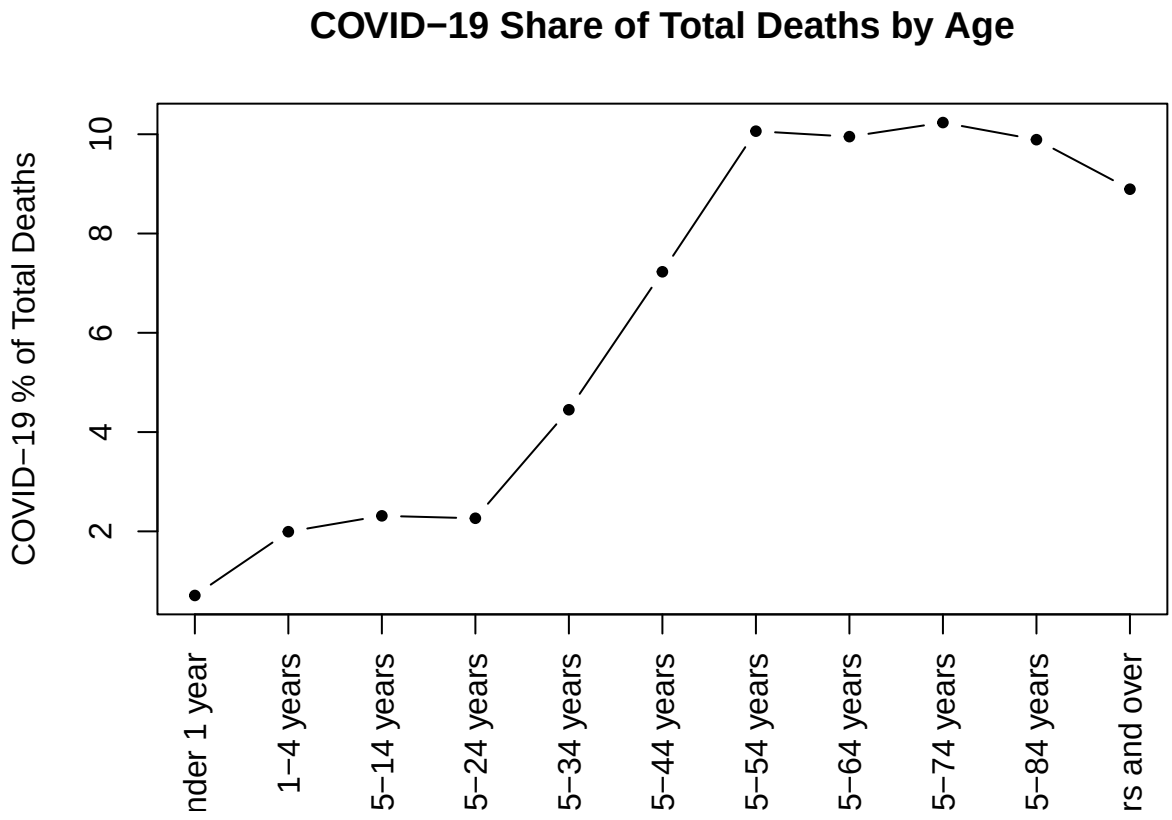
```
age_combined <- aggregate(
  cbind(COVID.19.Deaths, Total.Deaths) ~ Age.Group,
  data = covid_cleaned,
  sum
)

age_combined$Age.Group <- factor(
  age_combined$Age.Group,
  levels = c("Under 1 year", "1-4 years", "5-14 years", "15-24 years",
             "25-34 years", "35-44 years", "45-54 years", "55-64 years",
             "65-74 years", "75-84 years", "85 years and over"),
  ordered = TRUE
)
age_combined <- age_combined[order(age_combined$Age.Group), ]


pct <- (age_combined$COVID.19.Deaths / age_combined$Total.Deaths) * 100
```

```r
plot(
  pct,
  type = "b",
  pch = 20,
  xaxt = "n",
  xlab = '',
  ylab = "COVID-19 % of Total Deaths",
  main = "COVID-19 Share of Total Deaths by Age"
)

axis(1, at = 1:nrow(age_combined), labels = age_combined$Age.Group, las = 2)
```



This line plot shows the percentage of COVID-19 deaths relative to total deaths across the age groups. It portrays how the percentage of deaths change with the age groups.

Plot 5: COVID-19 vs. Pneumonia Share of Deaths by Age and Sex

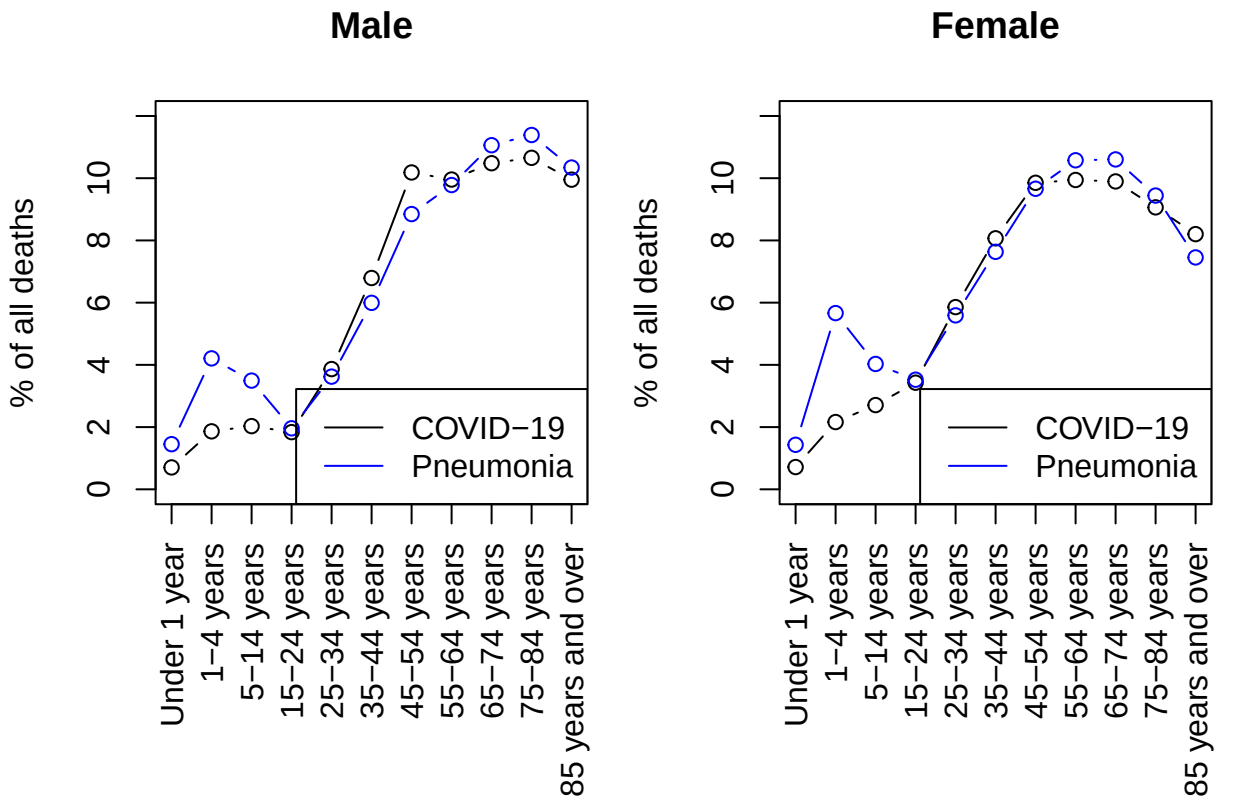
```r
male <- covid_cleaned[covid_cleaned$Sex == "Male", ]
female <- covid_cleaned[covid_cleaned$Sex == "Female", ]

par(mfrow = c(1, 2), mar = c(8, 4, 4, 1))

plot(male$covid_pct_of_all, type = "b", ylim = c(0, 12), xaxt = "n",
     xlab = "", ylab = "% of all deaths", main = "Male")
```

```r
lines(male$pneu_pct_of_all, type = "b", col = "blue")
axis(1, at = 1:11, labels = male$Age.Group, las = 2)
legend("bottomright", legend = c("COVID-19", "Pneumonia"),
       col = c("black", "blue"), lty = 1)


plot(female$covid_pct_of_all, type = "b", ylim = c(0, 12), xaxt = "n",
     xlab = "", ylab = "% of all deaths", main = "Female")
lines(female$pneu_pct_of_all, type = "b", col = "blue")
axis(1, at = 1:11, labels = female$Age.Group, las = 2)
legend("bottomright", legend = c("COVID-19", "Pneumonia"),
       col = c("black", "blue"), lty = 1)
```




```r
par(mfrow = c(1, 1))
```

These line plots compares how much of all deaths in each age group are due to COVID-19 versus pneumonia. Sex is shown separately to see whether age and sex change which illness accounts for more deaths.

Both COVID-19 and pneumonia related deaths increase as age increases, so age has a visible effect on deaths. Sex does not appear to have a large effect between them, as the two lines follow similar patterns. However, at ages 85 and older, the shares drop for females by about 2-3% in both COVID-19 and pneumonia while males stay near 10%.

Plot 6: Side-by-Side Bar Plot for Male vs. Female Covid-19 Deaths by Age Group

```
age_deaths_sex <- aggregate(
  COVID.19.Deaths ~ Age.Group + Sex,
  data = covid_cleaned,
  sum
)

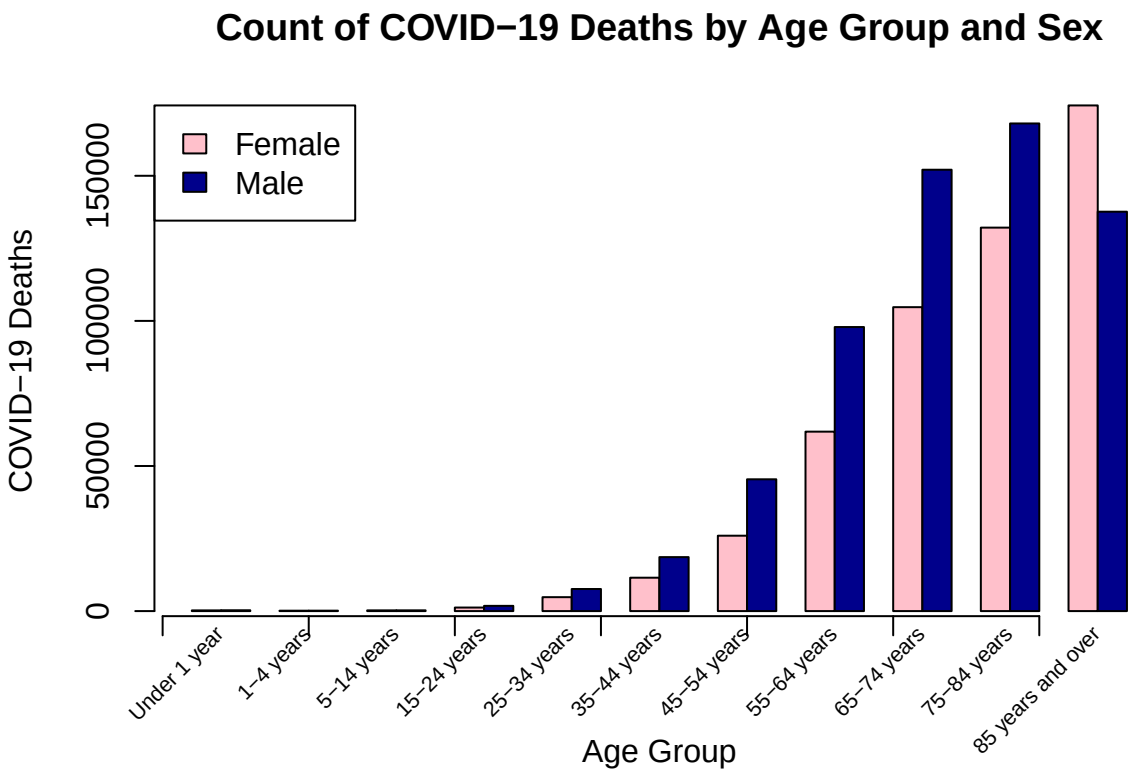
age_deaths_sex$Age.Group <- factor(
  age_deaths_sex$Age.Group,
  levels = c("Under 1 year", "1-4 years", "5-14 years", "15-24 years",
             "25-34 years", "35-44 years", "45-54 years", "55-64 years",
             "65-74 years", "75-84 years", "85 years and over"),
  ordered = TRUE
)

death_matrix <- xtabs(
  COVID.19.Deaths ~ Age.Group + Sex,
  data = age_deaths_sex
)

barplot_byside <- barplot(
  t(death_matrix),
  beside = TRUE,
  axisnames = FALSE,
  main = "Count of COVID-19 Deaths by Age Group and Sex",
  xlab = "Age Group",
  ylab = "COVID-19 Deaths",
  col = c("pink", "darkblue"),
  legend.text = TRUE,
  args.legend = list(x = "topleft")
)


axis(side = 1, labels = FALSE)

text(
  x = colMeans(barplot_byside),
  y = -7000,
  labels = levels(age_deaths_sex$Age.Group),
  srt = 45,
  adj = 1,
  xpd = TRUE,
  cex = 0.7
)
```



This side-by-side bar plot shows the count comparison of COVID-19 Deaths based on 11 different age groups while also differentiating between the counts for males vs females.

This shows a similar trend to Plot 5, with the count of COVID-19 deaths increasing as the age group increases. However, the differences in count between males and females shows that typically males have a significantly higher COVID-19 death count per age group until the oldest age group of 85 years and over, where females have a higher COVID death count than males.

Plot 7: Side-by-Side Bar Plot for Male vs. Female Pneumonia Deaths by Age Group

```
age_deaths_sex2 <- aggregate(
  Pneumonia.Deaths ~ Age.Group + Sex,
  data = covid_cleaned,
  sum
)

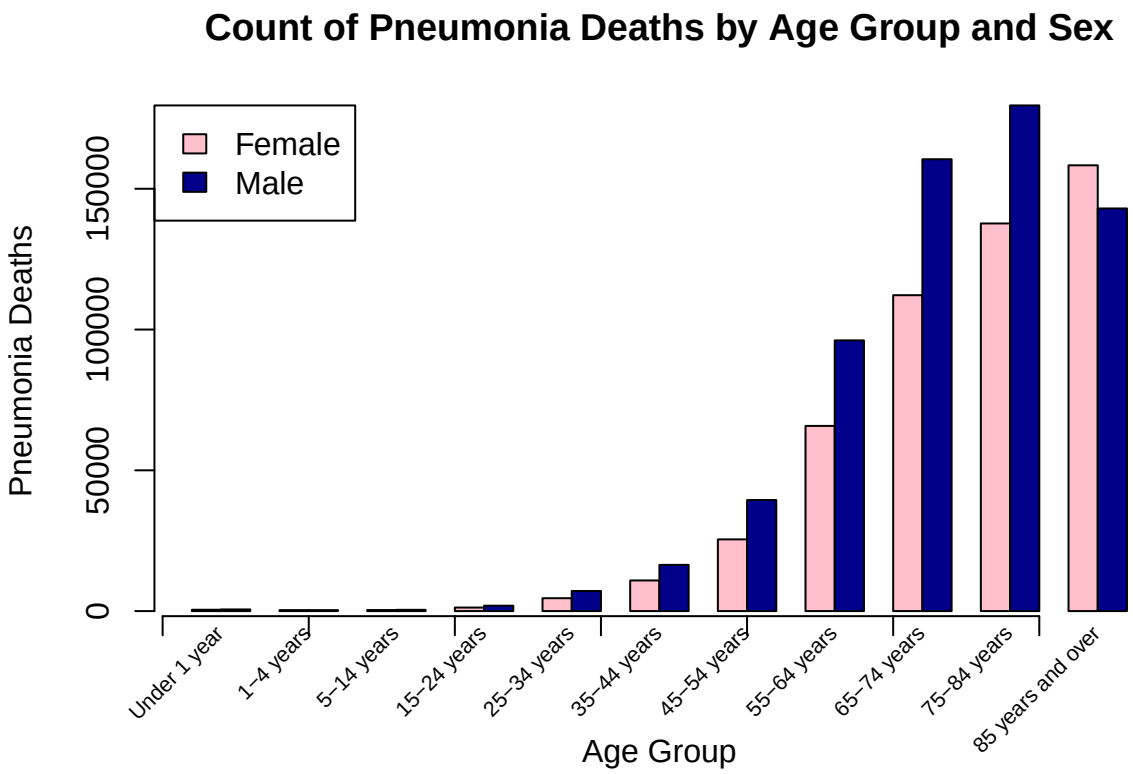
age_deaths_sex2$Age.Group <- factor(
  age_deaths_sex2$Age.Group,
  levels = c("Under 1 year", "1-4 years", "5-14 years", "15-24 years",
             "25-34 years", "35-44 years", "45-54 years", "55-64 years",
             "65-74 years", "75-84 years", "85 years and over"),
  ordered = TRUE
)
```

```r
death_matrix2 <- xtabs(
  Pneumonia.Deaths ~ Age.Group + Sex,
  data = age_deaths_sex2
)

barplot_byside2 <- barplot(
  t(death_matrix2),
  beside = TRUE,
  axisnames = FALSE,
  main = "Count of Pneumonia Deaths by Age Group and Sex",
  xlab = "Age Group",
  ylab = "Pneumonia Deaths",
  col = c("pink", "darkblue"),
  legend.text = TRUE,
  args.legend = list(x = "topleft")
)


axis(side = 1, labels = FALSE)

text(
  x = colMeans(barplot_byside2),
  y = -7000,
  labels = levels(age_deaths_sex2$Age.Group),
  srt = 45,
  adj = 1,
  xpd = TRUE,
  cex = 0.7
)
```



This side-by-side bar plot shows the count comparison of Pneumonia Deaths based on 11 different age groups while also differentiating between the counts for males vs females.

Similar to Plot 6, males again have a significantly higher Pneumonia death count per age group until the oldest age group of 85 years and over, where females have a higher death count than males. However, this higher female death count is less prominant than it was for COVID-19 deaths.

Plot 8: Side-by-Side Bar Plot for Male vs. Female Total Deaths by Age Group

```
age_deaths_sex3 <- aggregate(
  Total.Deaths ~ Age.Group + Sex,
  data = covid_cleaned,
  sum
)

age_deaths_sex3$Age.Group <- factor(
  age_deaths_sex3$Age.Group,
  levels = c("Under 1 year", "1-4 years", "5-14 years", "15-24 years",
             "25-34 years", "35-44 years", "45-54 years", "55-64 years",
             "65-74 years", "75-84 years", "85 years and over"),
  ordered = TRUE
)

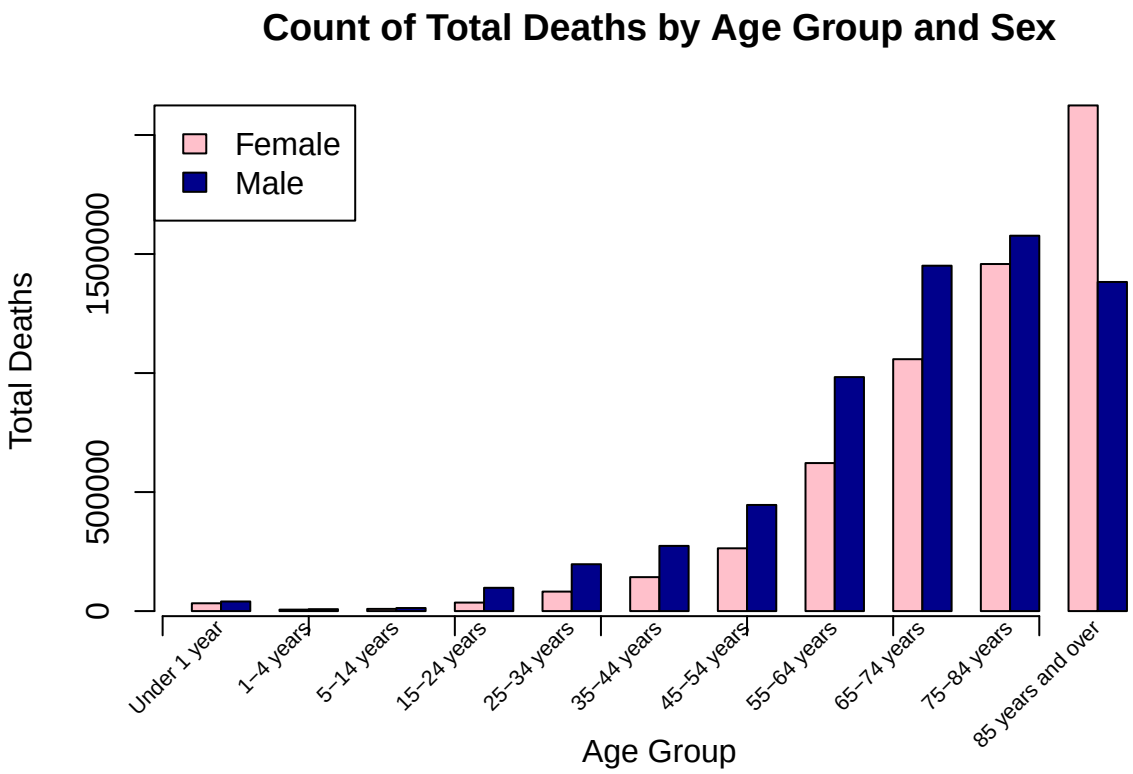
death_matrix3 <- xtabs(
  Total.Deaths ~ Age.Group + Sex,
```

```
  data = age_deaths_sex3
)

barplot_byside3 <- barplot(
  t(death_matrix3),
  beside = TRUE,
  axisnames = FALSE,
  main = "Count of Total Deaths by Age Group and Sex",
  xlab = "Age Group",
  ylab = "Total Deaths",
  col = c("pink", "darkblue"),
  legend.text = TRUE,
  args.legend = list(x = "topleft")
)


axis(side = 1, labels = FALSE)

text(
  x = colMeans(barplot_byside3),
  y = -60000,
  labels = levels(age_deaths_sex3$Age.Group),
  srt = 45,
  adj = 1,
  xpd = TRUE,
  cex = 0.7
)
```



This side-by-side bar plot shows the count comparison of Total Deaths based on 11 different age groups while also differentiating between the counts for males vs females.

This graph shows similar discrepancies between male and female deaths as Plots 5 and 6, but for the oldest age group the significantly higher female death count can larger be attributed to COVID-19 over Pneumonia based on the prior side-by-side bar plots.